Devansh Tripathi

Roll No.: IMS22090
Integrated BS-MS Programme
IISER Thiruvananthapuram

(+91)9580752916
devansh22@iisertvm.ac.in
GitHub Profile
Personal Website

Education

- **Indian Institute of Science Education and Research (IISER), Thiruvananthapuram** 2022 – Present
  Integrated BS-MS Programme, Applied Mathematics Major CGPA: 9.1/10
- **V.K.S Saraswati Vidya Niketan Inter College, Gola (Kheri)** 2008 – 2022
  Secondary and Higher Secondary Schooling, Mathematics and Computer Science

Research Experience

- **Universal Approximation Property of Foundation Models for Computational PDEs** May – Aug 2025
  ETH Zürich, Switzerland, Mentor: Prof. Dr. Siddhartha Mishra
  - Studied theoretical foundations of operator-learning foundation models for PDEs, including Convolution Neural Operator (CNO), POSEIDON, and Geometry Aware Operator Transformer (GAOT).
  - Investigated universal approximation property for GAOT as a PDE solution operator between L^p spaces.
  - Applied techniques from PDE analysis, functional analysis, measure theory, and Fourier analysis.

- **Distributed Memory Parallelization of Lax-Wendroff Flux Reconstruction** Jun – Jul 2024
  TIFR-CAM SSRP Program, Mentor: Prof. Praveen Chandrashekar
  - Parallelized Lax-Wendroff Flux Reconstruction methods for hyperbolic conservation laws using MPI.
  - Achieved 13x speedup with 82% efficiency on multicore systems.
  - Implemented Remote Memory Access (RMA) techniques for linear advection problems.
  - Contributed to open-source scientific computing repository TrixiLW.jl.

Selected Software & Repositories

- **PoissonSolver.jl (Code)**
  Julia, MUMPS, Parallel Computing
  - Developed a 2D Poisson equation solver in Julia supporting mixed boundary conditions (Dirichlet, Neumann, Periodic).
  - Integrated `MUMPS.jl` for parallel distributed-memory solving across multicore architectures.
  - Benchmarked parallel `MUMPS.jl` implementation against Block Thomas solver, analyzing speedup factors and memory consumption for varying grid resolutions.

- **High-Performance Computing (HPC) Implementations (Code)**
  C, MPI, CUDA, OpenMP
  - Wrote educational code for Message Passing Interface (MPI), OpenMP, CUDA for BLAS level 1,2 and 3 operations.
  - Conducted extensive scaling studies comparing CPU and multi-GPU (CUDA) implementations for large dense matrix-vector multiplication (up to $N = 32768$).

Presentation

- **Distributed memory parallelization of Lax-Wendroff Flux Reconstruction** Aug 2024
  TIFR-CAM, Bangalore
  - Presented my summer research work to Computational PDEs research group of Prof. Praveen Chandrashekar
  - Slides for the presentation can be found on this link.

Relevant Coursework

Advanced Coursework: Functional Analysis, Measure Theory, Sobolev Spaces and Elliptic Boundary Value Problems, Partial Differential Equations, Finite Element Methods, Numerical Analysis, Numerical Solutions of Differential Equations, Scientific Computing, Variational Methods and Control Theory, Applied Stochastic Analysis.

Technical Skills

Programming Languages: Julia, C, C++, Python, LaTeX

Parallel Computing & HPC: MPI, OpenMP, CUDA, OpenACC, High-Performance Computing

Libraries & Tools: NumPy, Pandas, Matplotlib, Git, Linux, VS Code, Command Line

Scientific Computing Frameworks: MUMPS, Trixi.jl, TrixiLW.jl

Certification: Advanced MPI Programming (ARCHER2 UK National Supercomputing Service)

Research Interests: Numerical Analysis of PDEs, Scientific Computing, High-Performance Computing

Awards & Fellowships

- **ETH4D Fellowship**, ETH Zürich — 2025
- **Summer Student Research Fellowship (SSRF)**, TIFR-CAM — 2024
- **INSPIRE Fellowship**, Government of India — 2022
  Awarded to top 1% students at the state level.
- **Award for Academic Excellence**, Government of Uttar Pradesh — 2022

References

- **Prof. Dr. Siddhartha Mishra**
  Full Professor of Mathematics, ETH Zürich
  siddhartha.mishra@sam.math.ethz.ch
- **Dr. K.R Arun**
  Associate Professor, IISER Thiruvananthapuram
  arun@iisertvm.ac.in
- **Prof. Praveen Chandrashekar**
  Professor, TIFR-CAM Bangalore
  praveen@tifrbng.res.in